

CAN (Controller Area Network) Bus

Mixed-Signal Modelling and Verification of a Multi-Node Communication Protocol in eSim

Circuit Simulation Project Proposal | Sourabh Kumar

1. Overview

This project models a complete CAN (Controller Area Network) communication bus in eSim as a mixed-signal system, combining an analog physical layer simulated in Ngspice with digital protocol controllers written in VHDL and integrated through eSim's NGHDL interface.

CAN is the standard automotive and industrial fieldbus defined by ISO 11898. It is chosen here because, unlike simpler protocols such as UART or SPI, it exercises the full range of eSim's mixed-signal capability: a differential analog physical layer with defined electrical thresholds, a bit-synchronised digital controller with no shared clock line, non-destructive bus arbitration that depends on a node reading back the analog level it is driving, and a fault-confinement mechanism that progressively removes a faulty node from the bus.

2. What will be modelled

Analog physical layer. Differential pair with 45 ohm driver on-resistance, dual 120 ohm termination, bus capacitance, a bias network holding the recessive level at 2.5 V, and a differential receiver centred between the 0.5 V recessive and 0.9 V dominant thresholds. A transmission-line model represents 220 m of twisted pair.

Digital controller. Bit timing with hard synchronisation and resynchronisation, bit stuffing and destuffing, CRC-15, frame assembly and decoding, arbitration, acknowledgement, error signalling and fault confinement.

System. Four controller instances share a single differential pair. When several nodes transmit simultaneously, arbitration is resolved bit by bit on the physical bus: a node transmitting a recessive level that reads back dominant has lost, and withdraws without corrupting the winner's frame.

3. Design parameters

These are derived rather than assumed. The propagation segment follows from a measured round-trip delay over the modelled 220 m bus.

Parameter	Value	Basis
Bit rate	125 kbit/s	design choice
Bit time	8.000 us	1 / bit rate
Time quanta per bit	16 (tq = 500 ns)	design choice
Controller clock	2.000 MHz	one tq per cycle
SYNC / PROP / PS1 / PS2	1 / 5 / 6 / 4 tq	PROP from measurement
Sample point	12 tq = 75.00%	> one bus round trip
Synchronisation jump width	4 tq	bounds oscillator drift
Maximum bus length	220 m	from PROP_SEG
Driver on-resistance	45.0 ohm	1.5 V at 33.333 mA
Termination	120 ohm x 2	ISO 11898-2
CRC polynomial	0x4599	ISO 11898-1

4. What will be demonstrated

Data transfer — a transmitting node's identifier, data length code and payload recovered correctly by receiving nodes across the analog bus.

Timing and synchronisation — 16 time quanta per bit, sample point at 75%, hard synchronisation and resynchronisation bounded by the synchronisation jump width.

Arbitration — four nodes transmitting simultaneously; the lowest identifier wins and the losers withdraw one bit apart, in the order their identifiers predict, without corrupting the winner's frame.

Error handling — CRC, form, stuff and bit errors detected; error frames transmitted; transmit and receive error counters; error-active, error-passive and bus-off states.

Fault confinement — a physical bus fault drives the affected node to bus-off while healthy nodes continue unaffected.

5. Scale of the design

The FOSSEE procedure notes that simple simulations with one or two components will not be accepted. This design comprises an analog section with differential driver stages, dual termination, bias network, bus capacitance, differential receiver and a transmission-line model; a digital section of ten VHDL modules covering bit timing, bit stuffing, CRC-15, frame generation, frame decoding, the transmit datapath, error frame generation, fault-confinement counters, the node wrapper and the top-level model; and a mixed-signal integration placing four NGHDL controller instances on one analog bus with ADC and DAC bridges at every boundary.

6. Deliverables

eSim project with schematic and simulation; VHDL source integrated via NGHDL; Ngspice netlists and subcircuit for the analog physical layer; an abstract documenting theory, method and results with Ngspice and Python plots; and a verification matrix listing every test and its measured result.

7. References

ISO 11898-1:2015, *Road vehicles — Controller area network (CAN) — Part 1: Data link layer and physical signalling*.

ISO 11898-2:2016, *Part 2: High-speed medium access unit*.

Bosch CAN Specification Version 2.0, Robert Bosch GmbH, 1991.